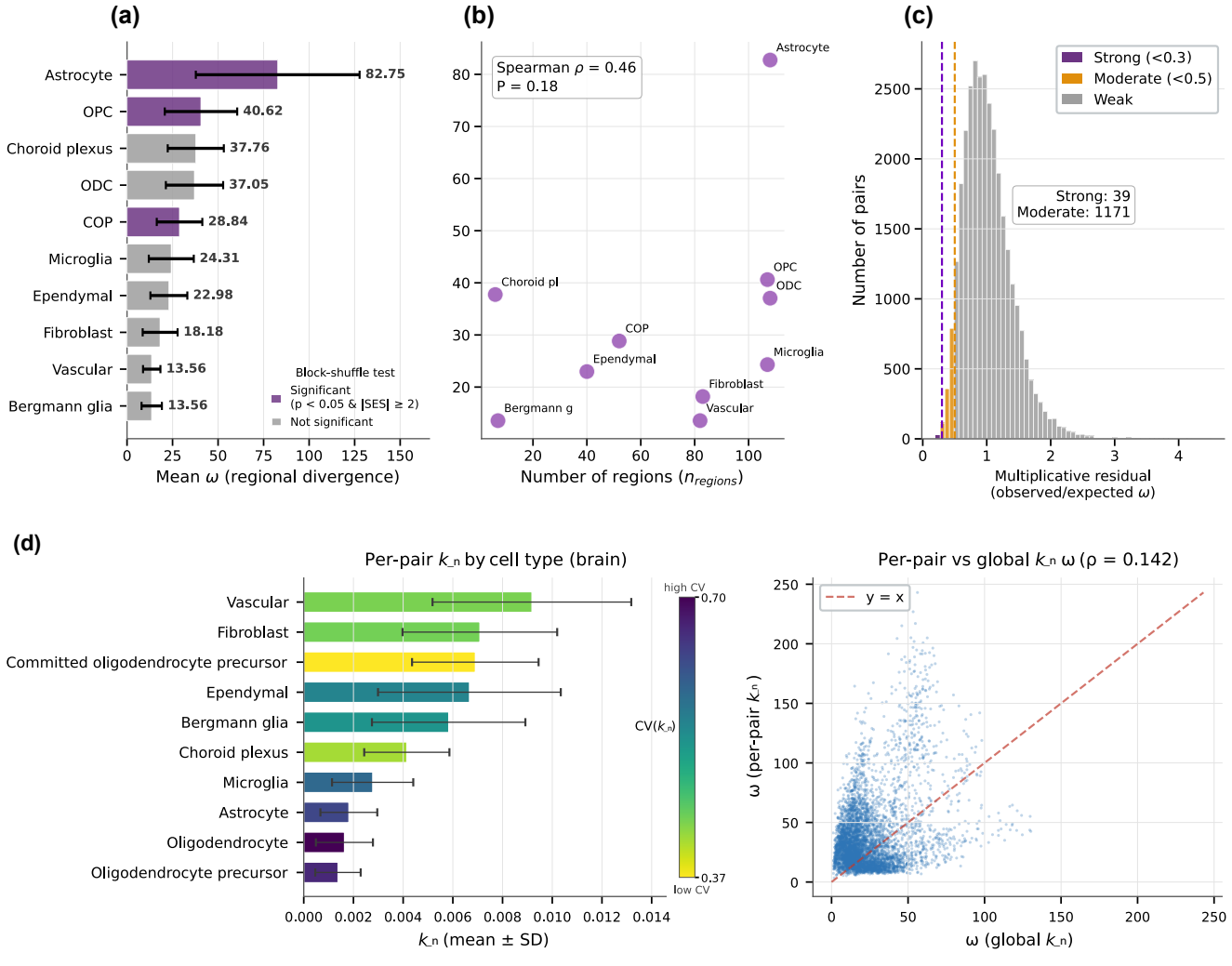

(a)
Astrocyte
OPC
Choroid plexus
ODC
COP
Microglia
Ependymal
Fibroblast
Vascular
Bergmann glia
82.75
40.62
37.76
37.05
28.84
24.31
22.98
18.18
13.56
13.56
Block-shuffle test
Significant
(p < 0.05 & |SES| ≥ 2)
Not significant
0
25
50
75
100
125
150
Mean ω (regional divergence)
(b)
Spearman ρ = 0.46
P = 0.18
Astrocyte
Choroid pl
OPC
ODC
COP
Ependymal
Microglia
Fibroblast
Bergmann g
Vascular
80
70
60
50
40
30
20
20
40
60
80
100
Number of regions ($n_{regions}$)
(c)
Strong (<0.3)
Moderate (<0.5)
Weak
Strong: 39
Moderate: 1171
Number of pairs
2500
2000
1500
1000
500
0
0
1
2
3
4
Multiplicative residual
(observed/expected ω)
(d)
Per-pair k_n by cell type (brain)
Vascular
Fibroblast
Committed oligodendrocyte precursor
Ependymal
Bergmann glia
Choroid plexus
Microglia
Astrocyte
Oligodendrocyte
Oligodendrocyte precursor
high CV
0.70
CV(k_n)
0.37
low CV
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
k_n (mean ± SD)
Per-pair vs global k_n ω (ρ = 0.142)
y = x
ω (per-pair k_n)
250
200
150
100
50
0
0
50
100
150
200
250
ω (global k_n)